

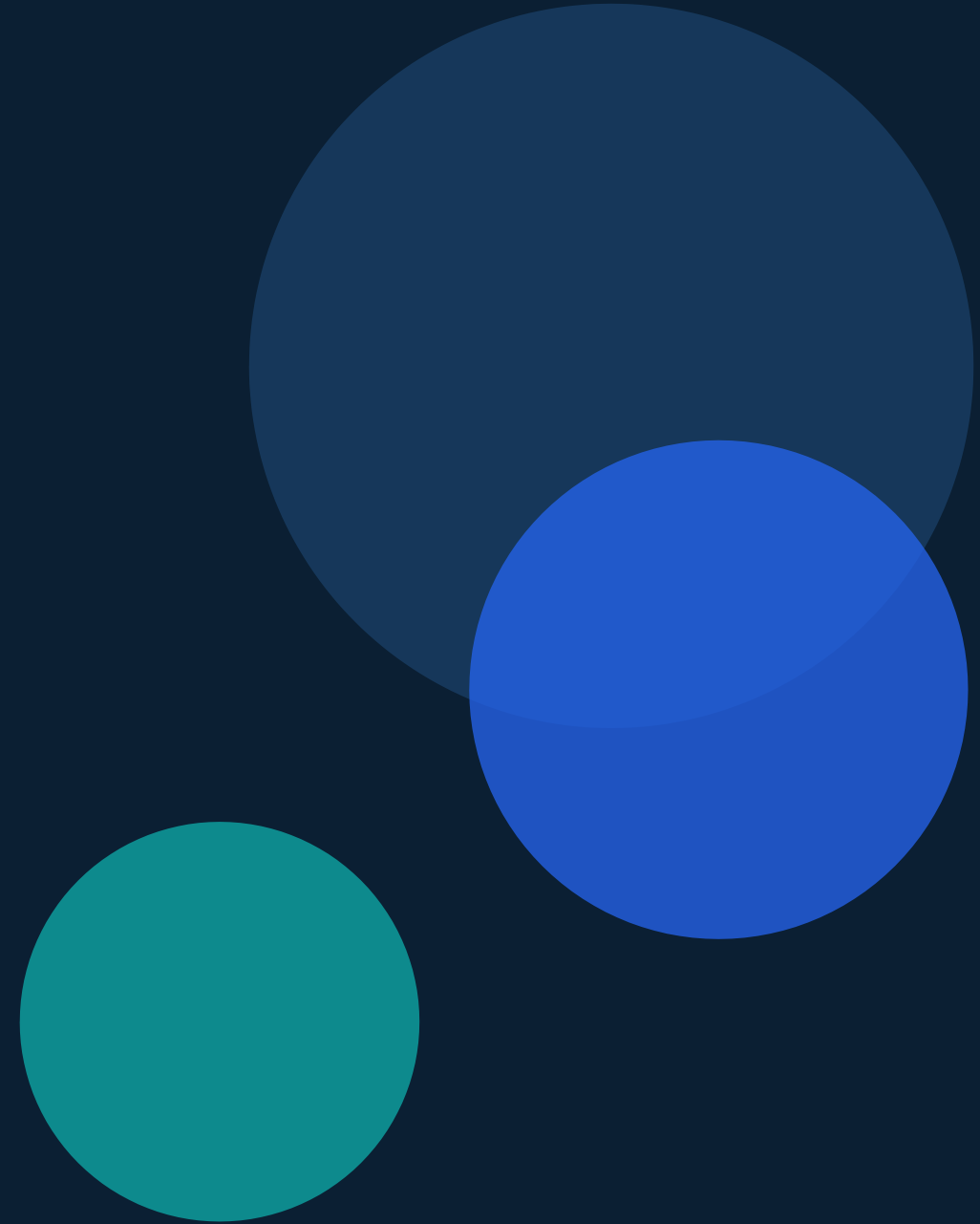
Meridian MMM + SHAP Explainability

A Bayesian analysis of channel contribution, saturation, ROI and future investment scenarios

Marketing Analytics Assignment

Prepared by Duaa Zahur

● Paid Search ● Shopping ● PMAX ● Video ● Facebook ● Instagram



Executive summary

The model points to a focused reallocation strategy, not a blanket budget increase.

KEY OUTCOMES

0.710

Holdout R²

Good generalization

5.41

Paid Search mROI

Best scale opportunity

63.5%

Meta contribution

Facebook + Instagram

+\$760.5k

Best profit change

Paid Search +50%



What the evidence says

Meta channels drive the largest volume, but Google Paid Search delivers the strongest financial efficiency and still has substantial headroom. PMAx and Shopping are closer to saturation, while Video is below break-even.



Recommended action

Scale Paid Search first, maintain controlled investment in Meta, trim PMAx and Shopping, and review Video. Avoid distributing incremental budget equally across all channels.

Assignment coverage and point alignment

All six required tasks are represented in the analysis and in this deck.

RUBRIC MAP

1

EDA

Missing values, spend trends, seasonality, outliers

10 pts



2

Data preparation

Daily modeling table, media, controls, train/validation

15 pts



3

Meridian MMM

Bayesian training, fit, contribution, adstock, saturation

25 pts



4

ROI & mROI

Channel efficiency, under/over-investment

20 pts



5

SHAP explainability

Summary, dependence, importance vs MMM

15 pts



6

Scenario simulation

Revenue, purchases, ROI and recommendations

15 pts



Data scope and modeling choice

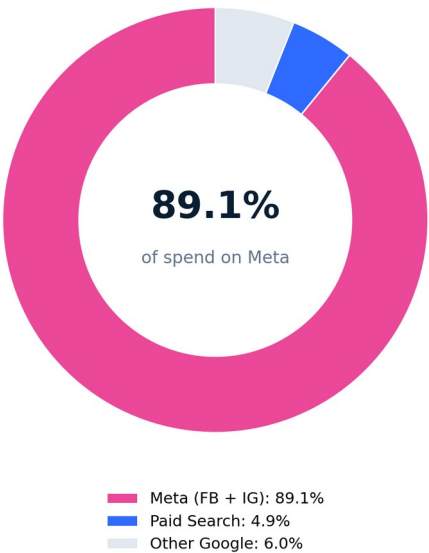
A single complete organization-level time series was selected for consistency and interpretability.

DATA SELECTION

Selected organization profile

Industry	Beauty & Fitness
Category	Face & Body Care
Currency	USD
Period	21 Nov 2020 - 19 May 2024
Observations	1,276 daily rows
Paid channels	6
Control variables	16

Historical Spend Concentration



Channels retained

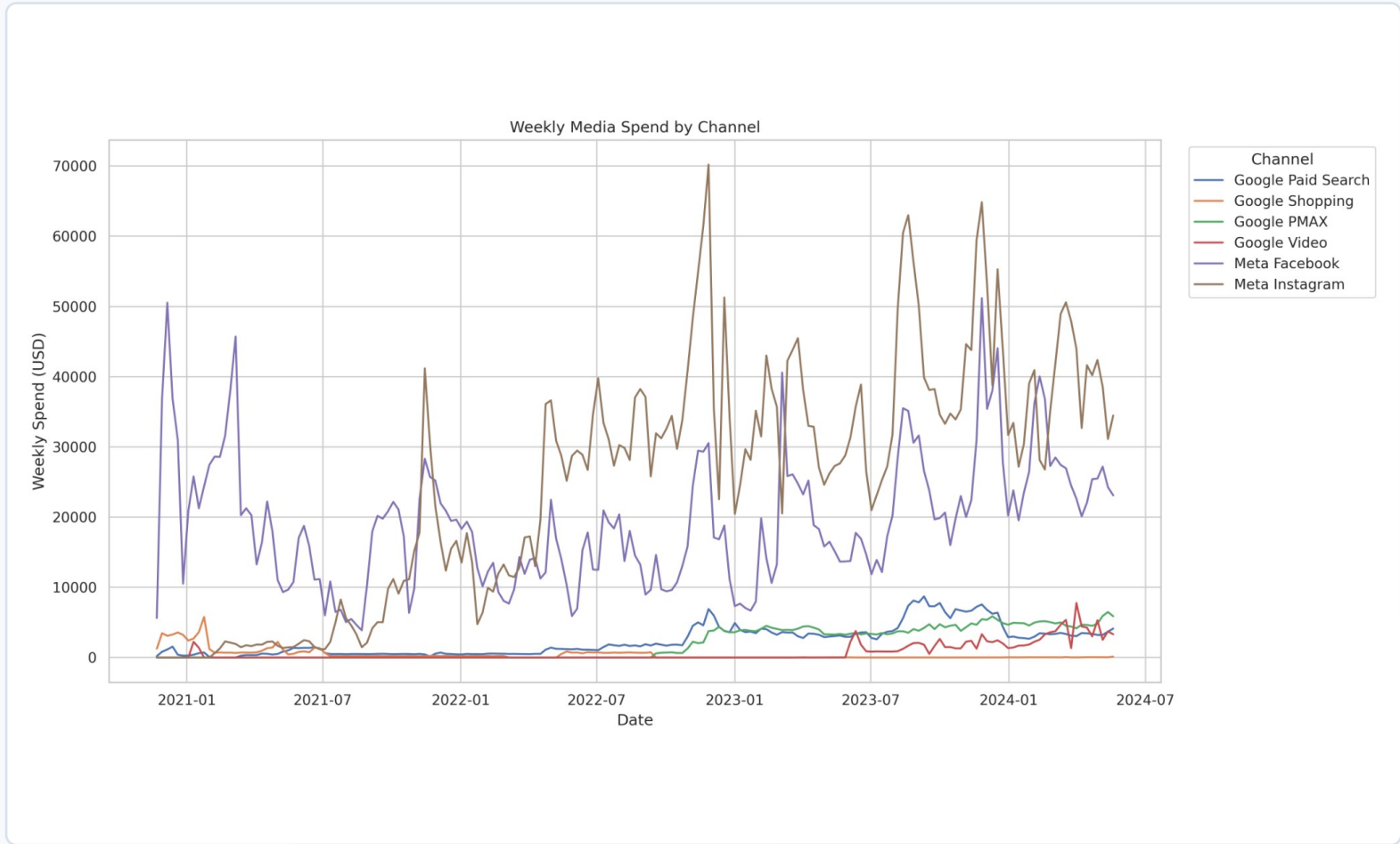
- Google Paid Search
- Google Shopping
- Google PMAx
- Google Video
- Meta Facebook
- Meta Instagram

Why one organization?
A complete, continuous daily series avoids mixing different currencies, business models and demand patterns.

Exploratory analysis

Seasonality and spend concentration are material features of this business.

TASK 1 · 10 POINTS



Spend concentration

Facebook and Instagram absorbed 89% of historical spend.



Seasonality

Sunday was the strongest weekday; November and Black Friday produced the largest spikes.



Outliers

15 purchase outliers were retained because they reflect real promotional demand.



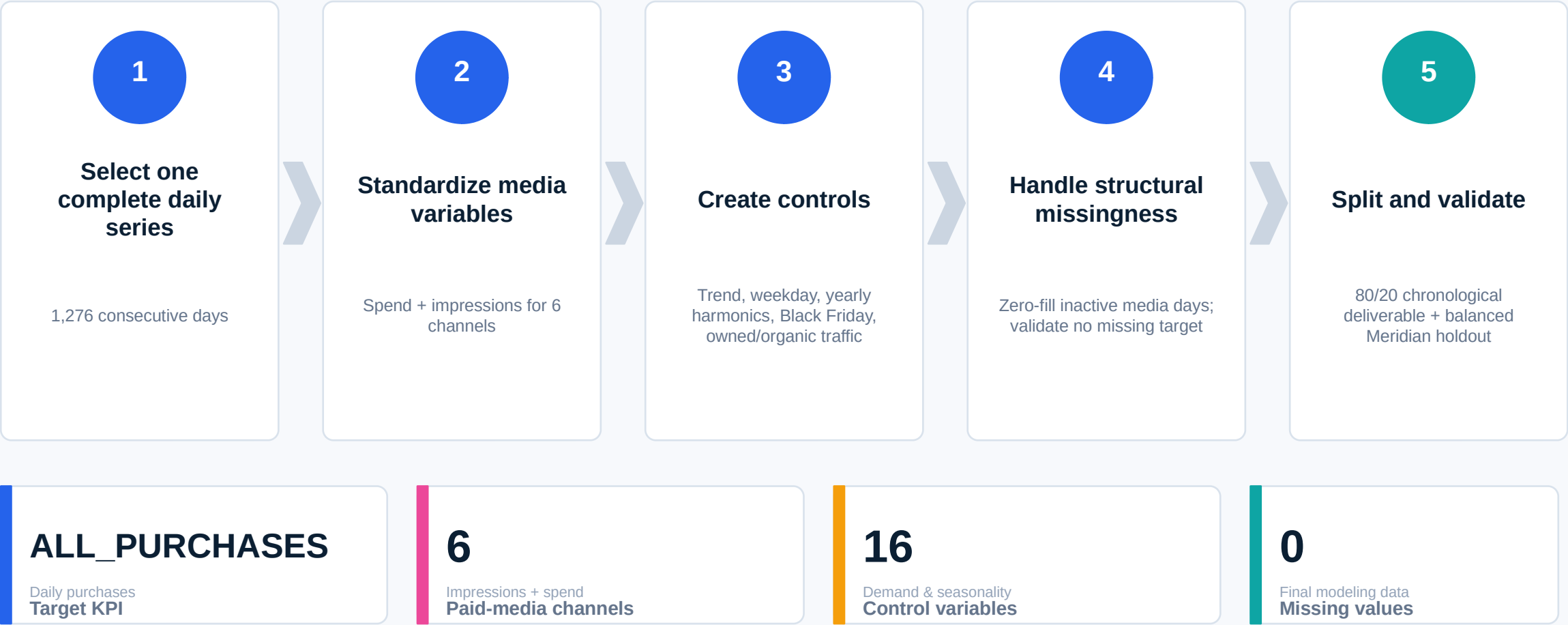
Trend

Purchases and media intensity increased meaningfully through 2023-24.

Data preparation and feature engineering

The final table links purchase demand to paid-media exposure, spend and demand controls.

TASK 2 · 15 POINTS



Meridian model design and training

The primary model is a Bayesian Marketing Mix Model trained with MCMC posterior sampling.

TASK 3 · 25 POINTS

Bayesian model workflow



Training step

``sample_posterior(...)`` performed the actual model fitting. Meridian does not use a scikit-learn ``.fit()`` call.



20%

Balanced holdout

Primary output

Contribution, adstock, saturation, ROI and response curves



4

MCMC chains



2,000

Posterior samples



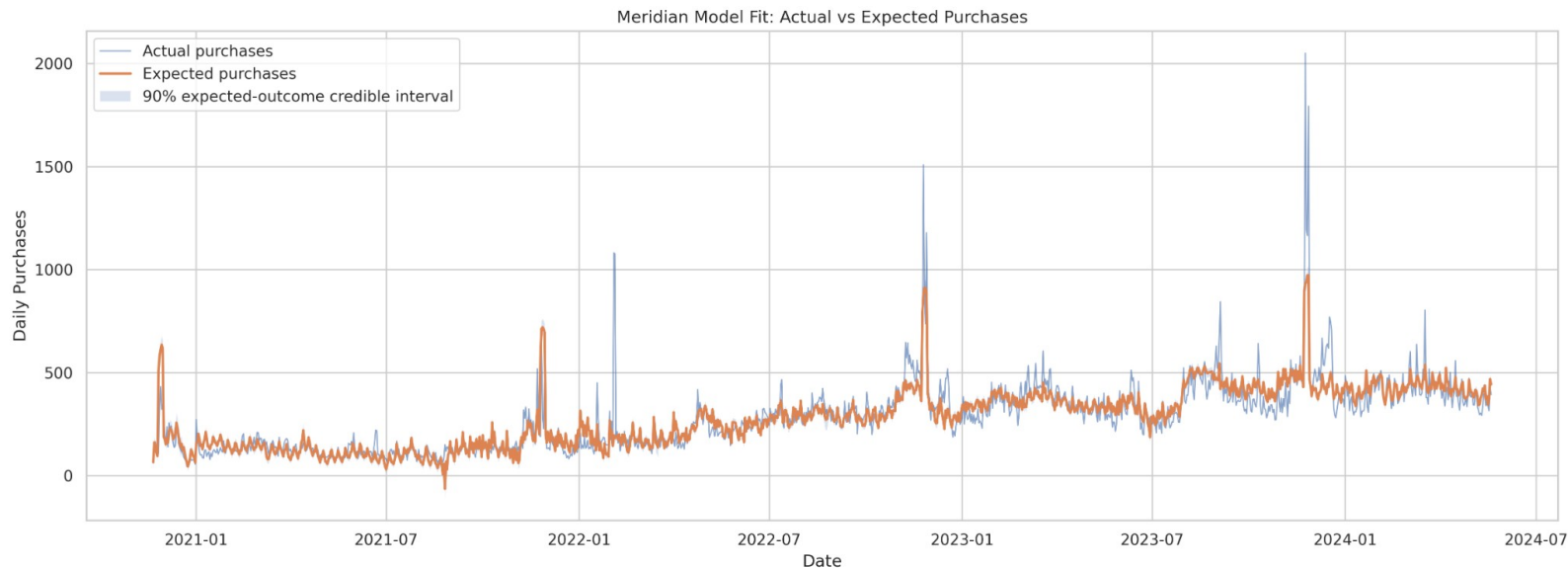
8 days

Maximum media lag

Model quality and holdout performance

The holdout metrics remain close to training results, indicating reasonable generalization.

MODEL VALIDATION



0.746

Training R^2

0.710

Holdout R^2

81.16

Purchases/day
Holdout RMSE

15.92%

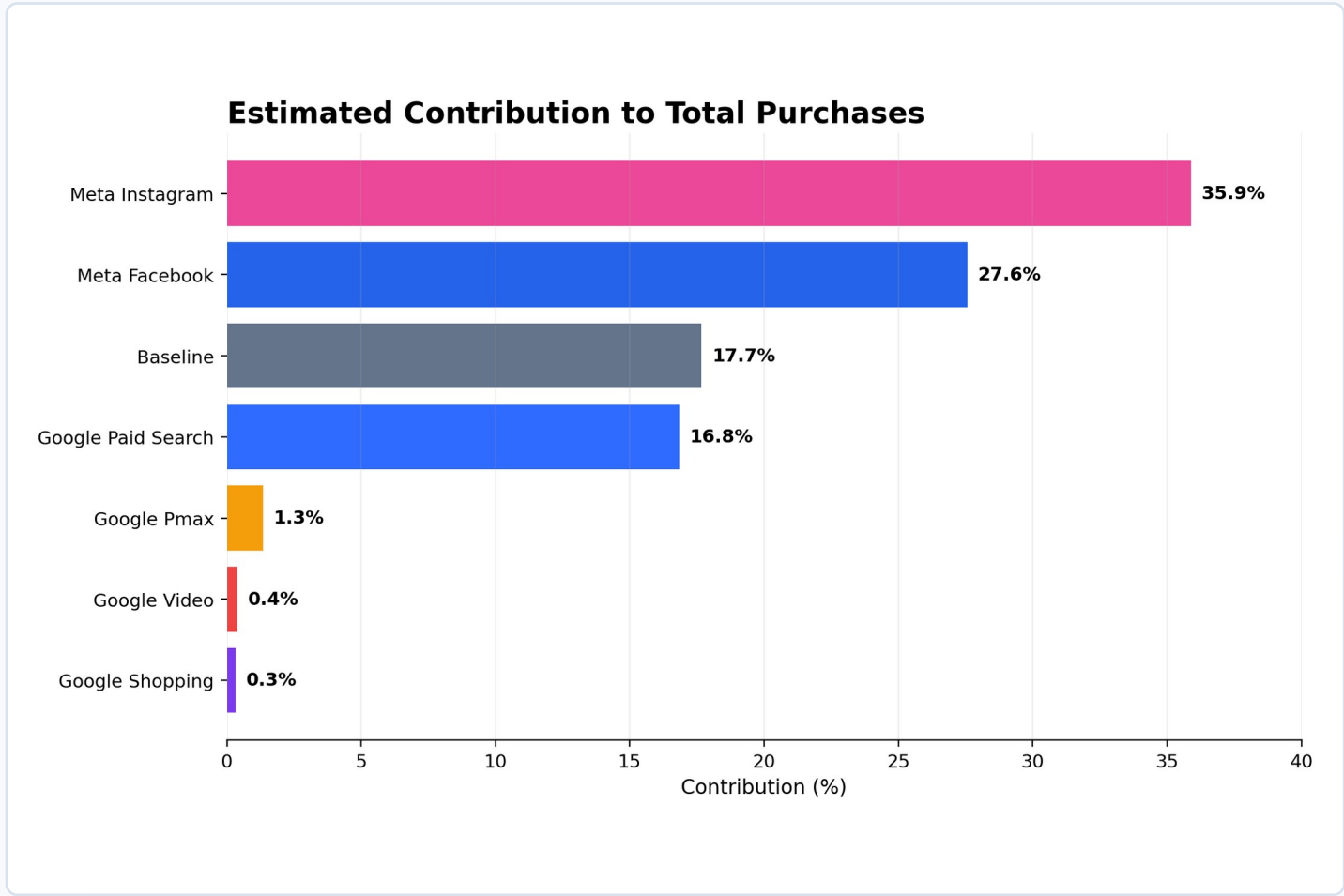
Holdout wMAPE

Interpretation: the small R^2 gap and slightly lower holdout error suggest limited overfitting. Extreme Black Friday peaks remain difficult to capture fully.

Channel contribution

Meta drives the most volume; Paid Search contributes far more than its budget share.

INCREMENTALITY



35.9%

Instagram

27.6%

Facebook

16.8%

Paid Search

17.7%

Baseline

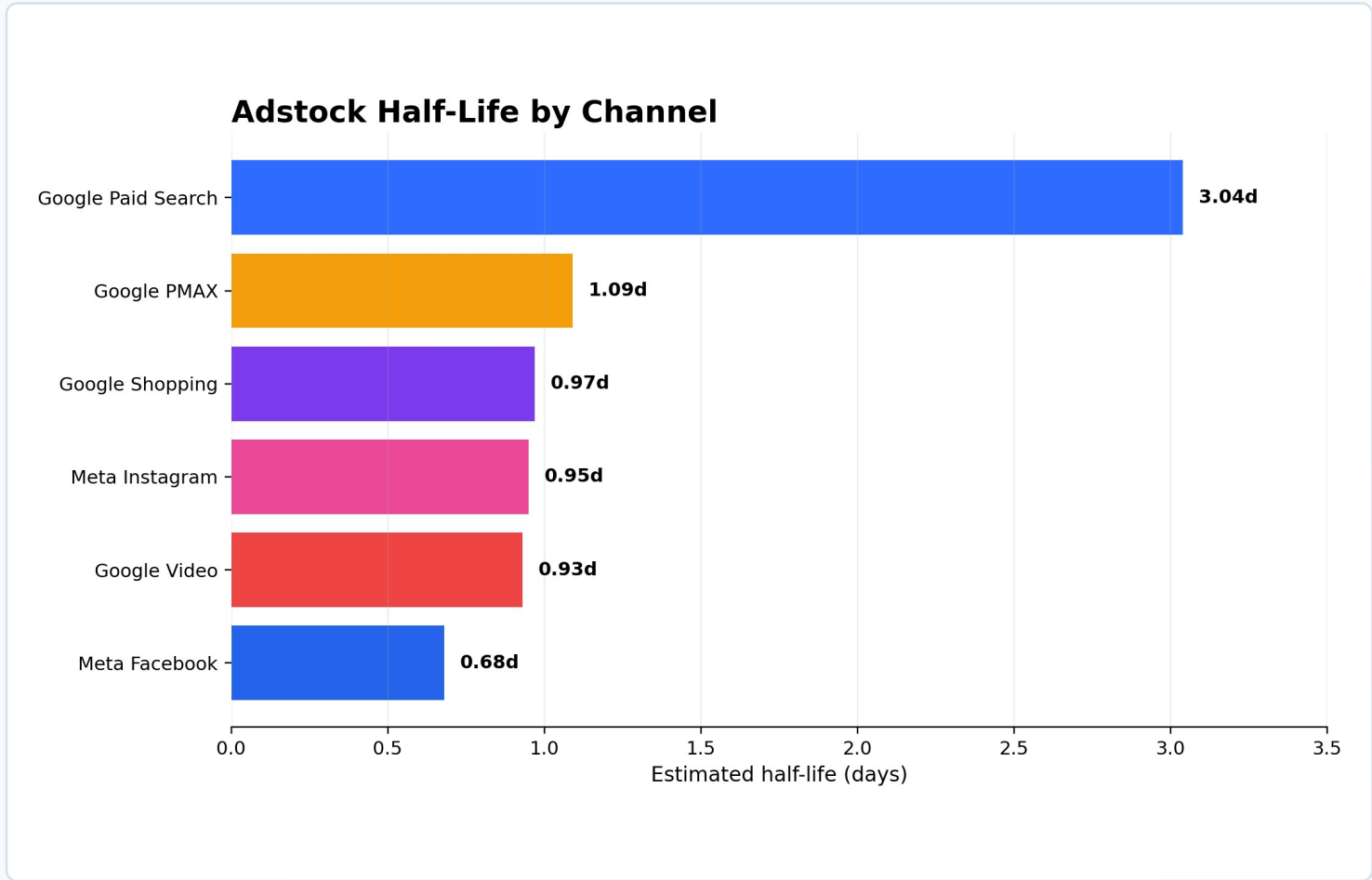
Core insight

Facebook and Instagram account for 63.5% of modeled purchases, broadly matching their dominant spend. Paid Search contributes 16.8% from only 4.9% of spend, a major efficiency signal.

Adstock and lagged media effects

Paid Search has a materially longer carryover window than the other channels.

CARRYOVER



Longest carryover

Google Paid Search half-life: 3.04 days; 35.2% of the initial effect remains after 7 days.



Fastest decay

Meta Facebook half-life: 0.68 days; less than 1% remains after 7 days.



Planning implication

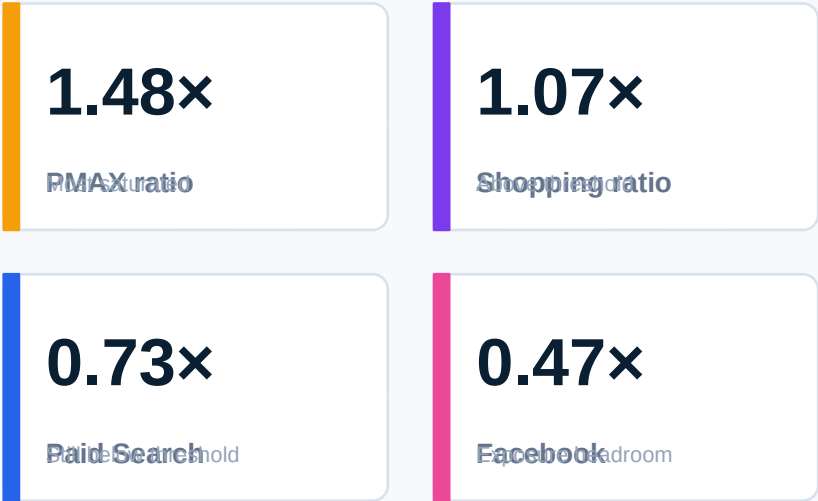
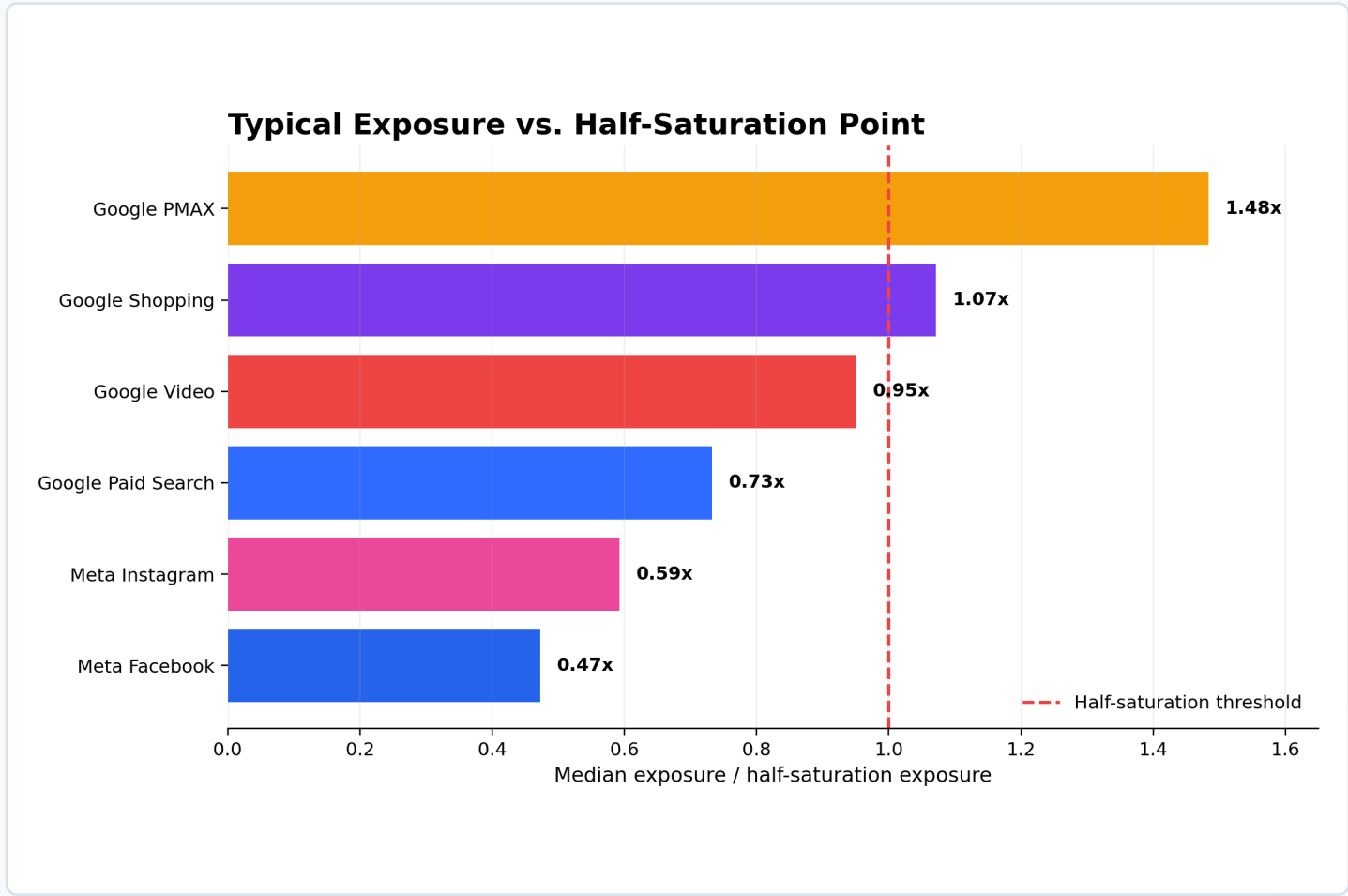
Search can influence purchases over several days. Meta requires more continuous exposure to sustain response.

Exact decay estimates have wide credible intervals for some channels and should be interpreted directionally.

Saturation and diminishing returns

PMAX and Shopping are operating closest to or beyond their half-saturation points.

HILL EFFECTS



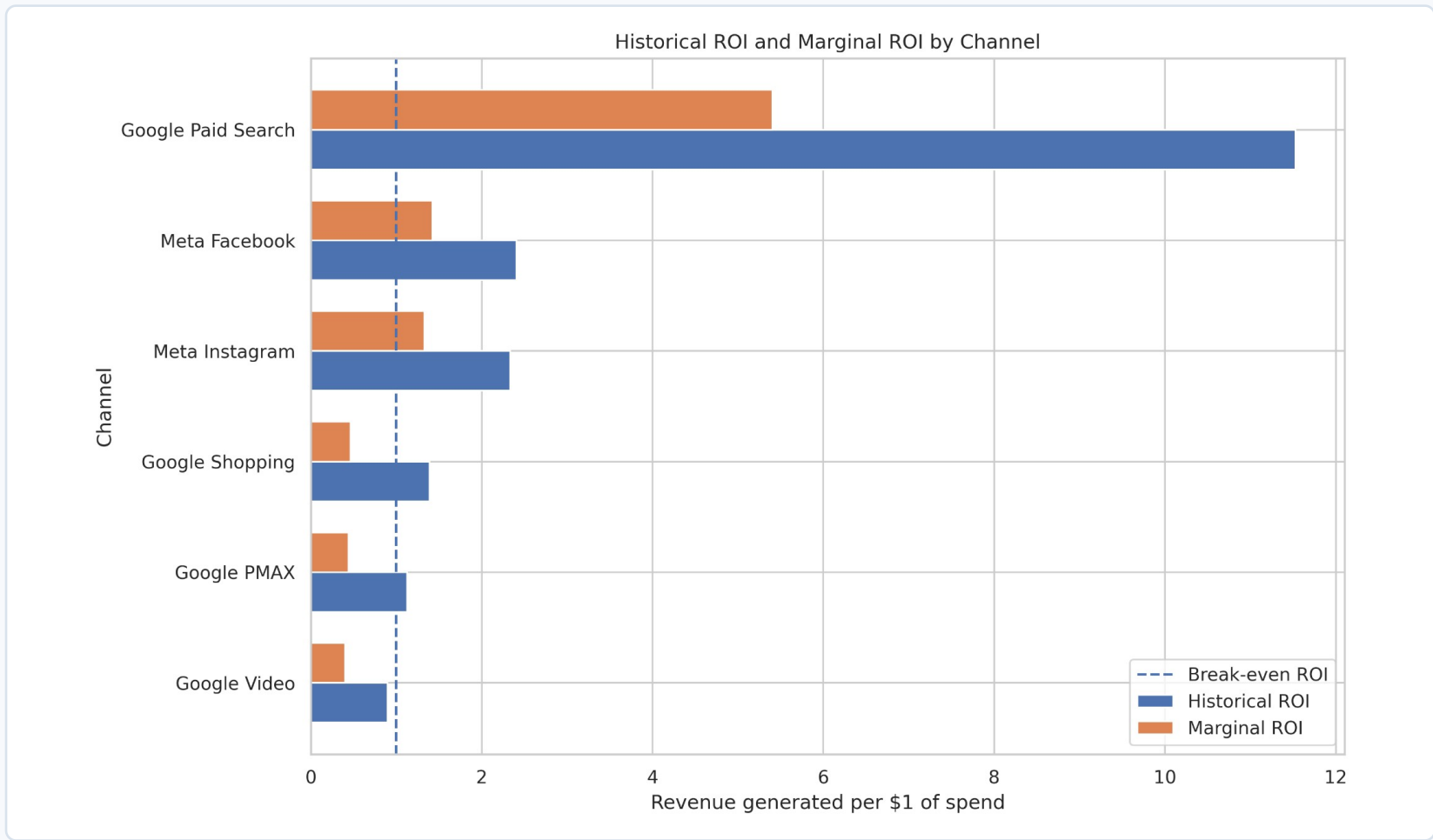
How to use this result

Saturation alone is not profitability. Final allocation decisions combine Hill headroom with ROI and marginal ROI. This is why Meta can show headroom but Paid Search remains the first scaling priority.

ROI and marginal ROI

Paid Search is the strongest scale opportunity; PMAx, Shopping and Video are weak at the margin.

TASK 4 · 20 POINTS



11.53 / 5.41

Paid Search ROI / mROI

2.40 / 1.42

Facebook ROI / mROI

2.33 / 1.33

Instagram ROI / mROI

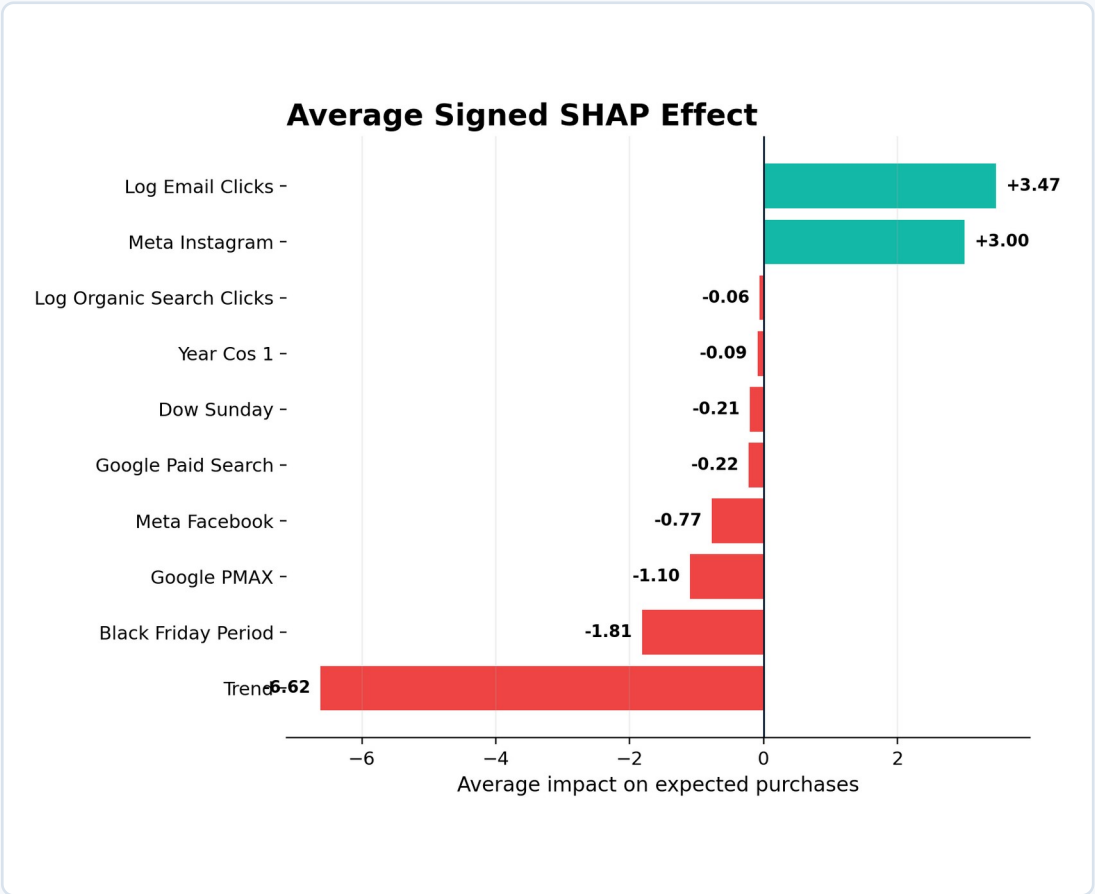
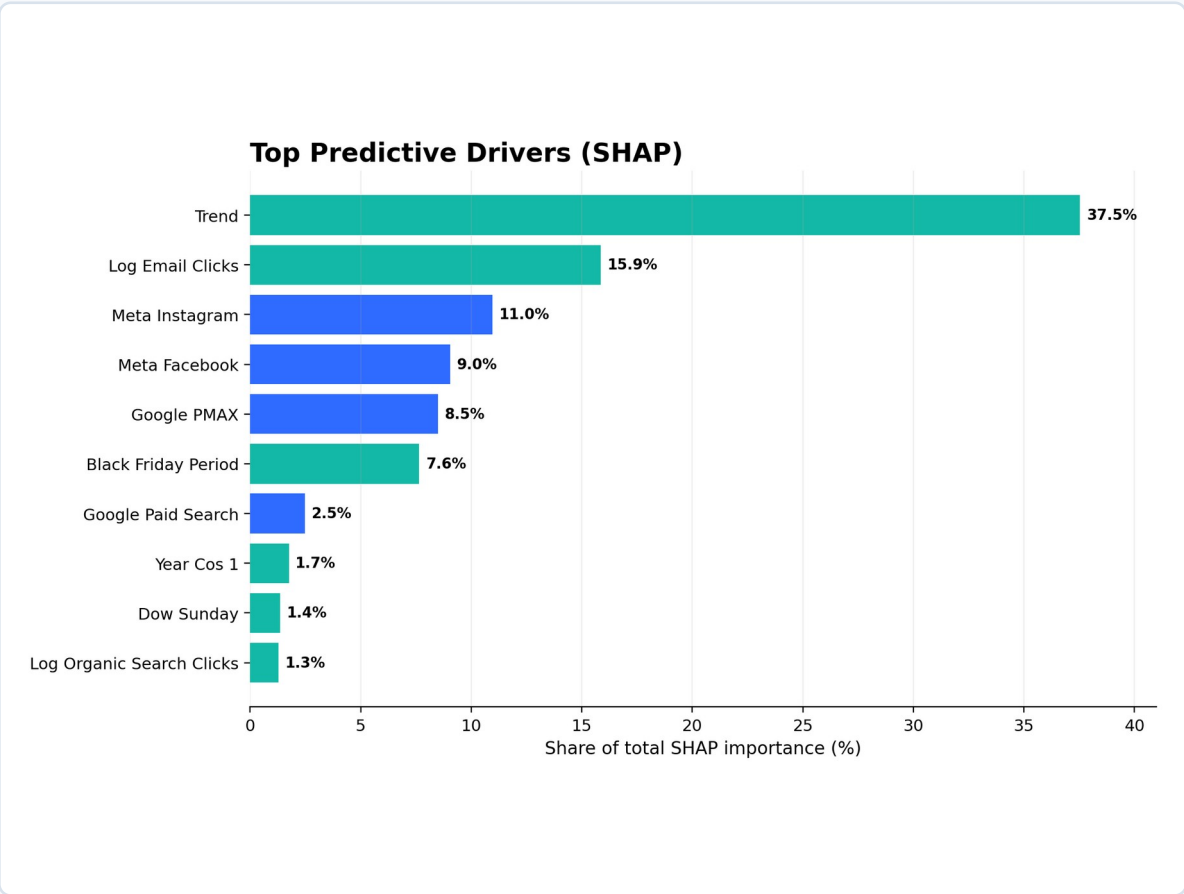
Decision rule

mROI > 1 supports incremental investment. Shopping (0.46), PMAx (0.44) and Video (0.40) should be held or reduced.

SHAP explainability

A high-fidelity Random Forest surrogate explains Meridian prediction patterns.

TASK 5 · 15 POINTS



0.974

Holdout

0.988

With Meridian

Surrogate R² Correlation

Interpretation

Trend and email dominate prediction; Meta and PMAX lead paid-media SHAP importance.

Important distinction

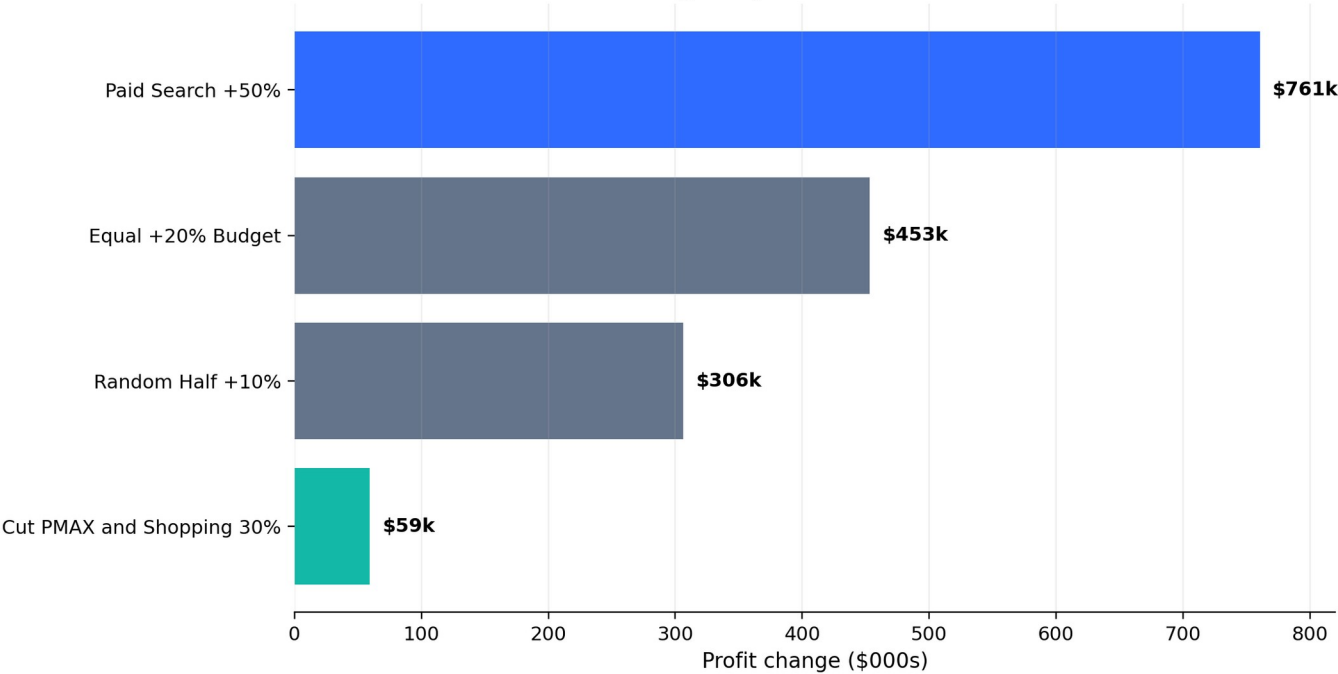
SHAP explains prediction patterns; Meridian remains the source for causal contribution, ROI and mROI.

Future marketing investment scenarios

Targeted reallocation outperforms broad-based spending increases.

TASK 6 · 15 POINTS

Estimated Profit Change by Scenario



Scenario comparison

Scenario	Profit change	Extra ROI
Paid Search +50%	+\$760.5k	4.46
Equal +20%	+\$453.3k	1.25
Random half +10%	+\$306.1k	1.60
Cut PMAx + Shopping	+\$59.3k	Cost saving

Best result

Paid Search +50% generated \$980k revenue and 11.6k purchases from \$220k extra spend.

Recommended next-quarter investment plan

Reallocate toward the channel with the strongest marginal economics.

BUSINESS RECOMMENDATION

SCALE

Google Paid Search

Increase investment first. Tested +50% scenario produced the strongest profit uplift and an extra-spend ROI of 4.46.

MAINTAIN / TEST

Meta Facebook + Instagram

Keep as major volume channels. Scale cautiously because they already represent 89% of spend and mROI is lower than Search.

REDUCE / HOLD

PMAX, Shopping, Video

Trim saturated PMAX and Shopping. Review Video because historical ROI and mROI are below break-even.

Best tested action: Paid Search +50% → +\$760.5k estimated profit | +\$980.4k revenue | +11,608 purchases

Monitor monthly

Conclusion, trust and limitations

The business answers are clear, but decisions should still be implemented gradually and measured.

FINAL ANSWERS

Five questions answered

- 1 Highest incremental purchases **Meta Instagram**
- 2 Strongest ROI and mROI **Google Paid Search**
- 3 Most saturated **Google PMAX**
- 4 Best tested allocation **Paid Search +50%**
- 5 Explainability **Strong surrogate fidelity; SHAP is predictive, Meridian is causal**

Limitations

- Results apply to one selected organization and period.
- MMM depends on causal assumptions and available control variables.
- Some channel estimates have wide credible intervals.
- SHAP explains a surrogate prediction model, not incrementality.
- Scenario results are estimates and should be validated with experiments.

Final recommendation Scale Paid Search, keep Meta disciplined, and remove low-return spend from saturated channels.